

# Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems

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## Abstract

Chart review in electronic health record (EHR) systems is rarely linear. A clinician investigating suspected septic shock may cycle through culture results, empirical antibiotic orders, lactate trends, and prior imaging within a single session—each pivot invalidating much of the retrieval context built up in the preceding turns. Standard session-based retrievers propagate stale query representations across these pivots, creating what we term *latent-context pollution*: embedding activations from a resolved topic degrade rankings for the newly active one.

We address this through **Geodesic Diagnostic Trajectories (GDT)**, which encodes evolving diagnostic intent as a trajectory on the Riemannian manifold  $\mathcal{S}_d^+$  of symmetric positive definite (SPD) matrices. The geodesic shift ratio  $\kappa_t$  quantifies intent displacement relative to recent trajectory pace; a soft-sigmoid gate controlled by  $\kappa_t$  suppresses stale context when a pivot is detected, with the degree of suppression subject to a formally proved interference bound (Proposition 1). A JEPA-style latent predictor [1, 2] anticipates next-turn information needs and populates a prefetch cache; prefetch volume is scaled by trajectory confidence to avoid flooding the cache after uncertain pivots [3].

We evaluated GDT against three comparators—CMA [4, 5], BM25 [6], and TF-IDF [7]—in a randomized four-arm crossover study spanning 124 sessions on 31 clinical vignettes drawn from MIMIC-III/IV [8, 9], eICU [10], and HiRID [11]. GDT reduced mean retrieval latency by 26.9% relative to TF-IDF ( $p < 0.001$ , Co-

hen’s  $d=-4.12$ ) and by 4.0% relative to CMA ( $p<0.001$ ). Proposition 1 held without exception over all 1,074 gate-triggered transitions. An M/M/c queueing analysis [12] translates the latency savings to an estimated 4.1-minute reduction in per-reviewer patient processing time during a peak-acuity 4-hour shift. Notably, all four arms produced 0% task completion; sparse lexical retrieval cannot bridge the vocabulary mismatch inherent to complex multi-pivot queries, a ceiling that GDT’s geometric mechanisms alone cannot overcome.

*Keywords:* Electronic health records, Riemannian manifolds, SPD matrices, geodesic shift ratio, differential geometry, clinical decision support, session-based retrieval, JEPA, optimal control, queueing theory.

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## 1. Introduction

The information-seeking behavior of clinicians during EHR chart review does not resemble the stable-query model assumed by most retrieval systems. A physician managing a post-surgical patient in the ICU may open a chart to check creatinine trends, notice an unexpected drop in platelet count, pivot to hematology consult notes, then jump to the anticoagulation record—all within a few minutes [13]. Each pivot erases much of the relevance of the previous retrieval context, yet standard session-based systems treat history as a smoothly accumulating buffer that grows monotonically helpful over time.

This assumption fails in two distinct ways. First, when a clinician abruptly changes topic, residual embedding activations and term weights from the prior topic degrade ranking quality for the new one—what we call *stale-context leakage*. Second, because each query is handled synchronously, there is no mechanism to exploit the fact that certain diagnostic transitions are predictable: a clinician reviewing sepsis markers is far more likely to next query vasopressor dosing than orthopedics.

Riemannian geometry offers a principled solution to both problems. Encoding session intent as a trajectory on the manifold  $\mathcal{S}_d^+$  of symmetric positive definite matrices provides a metric space in which geodesic distance naturally

measures how much the clinician’s information need has shifted between consecutive turns [14]. A large displacement on this manifold indicates an abrupt pivot; a small one indicates within-topic refinement. These two cases call for opposite responses from a retrieval system.

This paper introduces **Geodesic Diagnostic Trajectories (GDT)**, which operationalizes these geometric intuitions into four concrete mechanisms:

1. Session intent is represented as a trajectory  $\{S_t\}$  on  $\mathcal{S}_d^+$ , where each  $S_t$  is a regularized covariance over recent query embeddings.
2. A geodesic shift ratio  $\kappa_t$  drives a soft gate that suppresses stale history proportionally to detected pivot magnitude; the interference reduction is formally bounded (Proposition 1).
3. A JEPA-style latent predictor [1] forecasts next-turn intent and pre-populates the result cache; confidence scaling by  $\kappa_t$  prevents noisy prefetch after uncertain pivots.
4. An M/M/c queueing model [12] connects sub-second latency gains to concrete estimates of per-reviewer throughput improvement during peak-acuity periods.

We compare GDT against CMA [4, 5], BM25 [6], and TF-IDF [7] in a randomized four-arm within-subject crossover study on 31 complex multi-pivot clinical vignettes, discussing both the latency improvements that GDT achieves and the fundamental limitation—uniform 0% task completion—that identifies dense retrieval as the unavoidable next step.

### Statement of Significance

- **Problem or Issue:** EHR chart review involves frequent, abrupt topic shifts. Standard session-based retrievers suffer from “latent-context pollution” when stale query history degrades new search results.
- **What is Already Known:** Session intent modeling usually relies on monotonic accumulation of history, treating past queries as uniformly helpful.

- **What this Paper Adds:** Geodesic Diagnostic Trajectories (GDT) tracks intent on a Riemannian manifold, formally gating stale context via a geodesic shift ratio, and intelligently prefetching results using a JEPA-style predictor to cut latency by 27%.
- **Who Would Benefit:** Clinicians navigating complex patient charts and informaticians seeking to optimize operational EHR throughput and reduce cognitive load.

## 2. Background and Related Work

The starting point for our geometric machinery is the affine-invariant metric on  $\mathcal{S}_d^+$  due to Pennec et al. [14], developed in the context of computational anatomy where SPD matrices naturally arise as diffusion tensors in brain imaging. That metric is geometrically clean but computationally expensive; Arsigny et al. [15] proposed the Log-Euclidean approximation, which maps SPD matrices to a vector space via the matrix logarithm, enabling standard Euclidean operations at the cost of some curvature fidelity. Both frameworks have since found homes in deep learning: Bronstein et al. [16, 17] surveyed the broader manifold-valued representation learning effort, from which SPDNet [18] and related architectures [19] emerged. GDT borrows the covariance-as-intent-state idea from this lineage but applies it to a domain—EHR session context—where the geometry is an engineering choice rather than a physical constraint.

### 2.1. Self-Supervised Latent Prediction

LeCun’s JEPA framework [2] is, at its core, a rejection of the idea that useful representations must be learned by reconstructing raw pixels or tokens. The argument is simple: predicting in a compressed latent space forces the model to learn semantically meaningful structure while ignoring irreducible noise. That insight had precursors—Contrastive Predictive Coding [20] achieved strong downstream performance without reconstruction—but JEPA made the

generative-model-free position explicit and principled. Image and video instantiations (I-JEPA [1], V-JEPA [21], and their successors [22, 23]) validated the approach at scale. Cui et al. [24] later showed theoretically that JEPA’s latent energy minimization bounds predictive risk under distribution shift—a desirable property for GDT’s intent predictor, since diagnostic query distributions change continuously within and across sessions. Training uses a momentum encoder [25] to stabilize targets without explicit negative sampling.

## 2.2. Session-Based Search and Multi-Turn Drift

Sequential recommendation has a longer history than conversational search, and the two lines only partially converged. GRU4Rec [26] established that recurrent networks could capture within-session item dependencies; SASRec [27] and SR-GNN [28] later showed that self-attention and graph neural networks extracted richer transition patterns. In information retrieval the challenge is different: the query, not the item, changes turn-by-turn, requiring query rewriting [29] and multi-turn context modeling [30, 31, 32]. Dense bi-encoders such as DPR [33] and ANCE [34] handle the semantic gap that lexical methods cannot, and ColBERT [35] and ColBERTv2 [36] extend late-interaction scoring to multi-turn settings. Our own earlier systems—CMA [4, 5] and Semantic Intent Weaver [37]—showed that intent continuity could be tracked across turns, but relied on Euclidean distance heuristics that cannot distinguish topically stable refinement from abrupt pivot. GDT replaces those heuristics with geodesic displacement on  $\mathcal{S}_d^+$ .

## 2.3. EHR Retrieval and Clinical Foundation Models

Clinical NLP spent its first decade largely on rule-based extraction and statistical term weighting [38]; Shickel et al. [39] surveyed the transition to deep learning and made clear that raw text locked inside clinical notes was a mostly untapped resource. Sequential event models like Doctor AI [40] and Deep Patient [41] then shifted the focus to structured EHR timelines, encoding patient trajectories rather than individual documents. The emergence of transformer

pretraining changed both branches at once. ClinicalBERT [42] showed that even modest domain-adaptive fine-tuning of BERT on MIMIC notes moved the needle on clinical NER and relation extraction; BioBERT [43], BEHRT [44], and Med-BERT [45] each emphasized a different slice of the vocabulary–task space. At hospital scale, NYU-Tron [46] demonstrated that models trained on millions of real inpatient notes can match or exceed specialist predictions on readmission; Med-PaLM [47] and subsequent reasoning-tuned systems [48] pushed further into multi-step clinical question answering. The practical ceiling for these models turns out to be latency as much as accuracy [49]. EHRMamba [50], PORTER [51], and EHR-RAGp [52] each propose architectural routes to faster or more retrieval-aware inference; clinical RAG pipelines [53] attempt to decouple parametric knowledge from retrieved context. None of these systems, however, addresses what happens to retrieval quality when the clinician’s active question abruptly pivots mid-session—the precise scenario GDT is designed for.

#### 2.4. *Clinical Safety and Cognitive Governance*

The literature on clinical decision support has for decades documented a paradox: faster, more prominent suggestions tend to *increase* certain error modes rather than reduce them. Croskerry’s [54] analysis of premature closure—the tendency to stop gathering evidence once a plausible diagnosis is found—sits at one end of this spectrum; Goddard et al. [55] and others [56, 57] have catalogued how over-reliance on algorithmic recommendations creates a secondary anchoring problem. When a retrieval system prefetches documents and surfaces them before the clinician has finished processing the current page, the risk is not that the documents are wrong—they may well be correct—but that they narrow the cognitive search space prematurely [3]. GDT acknowledges this tension directly. The confidence-score display and source provenance labels are not cosmetic additions; they are intended to signal to the clinician when the system’s own uncertainty is high and the suggestion should be treated skeptically. The reduced prefetch volume after uncertain pivots serves the same purpose from the supply side: if the model does not know where the diagnostic trajectory is

heading, it sends fewer unsolicited suggestions rather than more. Any production deployment would additionally require formal CONSORT [58] or PRISMA [59] trial reporting, usability validation via NASA-TLX [60] or SUS [61], and SMART on FHIR [62] integration for interoperability. Ongoing monitoring for population-level distributional drift [63], subgroup fairness [64], and explanation quality [65] are prerequisites for sustained responsible use [66, 67].

### 2.5. Novelty

Taken individually, the building blocks of GDT—SPD covariance encoding, geodesic distance, JEPA-style prefetch, queueing-theoretic projection—each have precedents in other domains. What is new is the combination and the clinical grounding. No existing retrieval framework ties Riemannian intent tracking to a formally proved stale-context interference bound, gates history reuse by a normalized geodesic acceleration measure, and scales prefetch aggressiveness by the predictor’s own trajectory confidence. Adding the M/M/c throughput model then connects these millisecond-level optimizations to ward-level operational outcomes. GDT is, to our knowledge, the first system to do all four together, and to evaluate the resulting pipeline on a multi-database clinical benchmark with an explicit adversarial multi-pivot structure.

## 3. The GDT Framework

### 3.1. Diagnostic State Space

**Definition 1** (Diagnostic State Space). *The diagnostic state space is  $\mathcal{M} = \mathcal{S}_d^+$ , the Riemannian manifold of  $d \times d$  symmetric positive definite matrices, equipped with the affine-invariant metric tensor*

$$g_S(u, v) = \text{Tr}(S^{-1}u S^{-1}v), \quad S \in \mathcal{S}_d^+, \quad u, v \in T_S \mathcal{S}_d^+. \quad (1)$$

*The geodesic distance between two SPD points is  $d_g(S_a, S_b) = \|\log(S_a^{-1/2} S_b S_a^{-1/2})\|_F$ .*

### 3.2. SPD Intent Representation

Each query  $q_t$  is encoded by ClinicalBERT and projected to  $d=128$  dimensions, giving an embedding  $e_t \in \mathbb{R}^d$ . The local intent matrix summarizes the recent  $k$ -turn context as a regularized sliding-window covariance:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d. \quad (2)$$

Projecting to  $d=128$  is a deliberate trade-off: eigendecomposition costs  $\mathcal{O}(d^3) \approx 2.1 \times 10^6$  FLOPs per turn, which we measured to be under 5% of total per-turn computation.

### 3.3. Geodesic Shift Ratio

Rather than thresholding raw geodesic distance—which varies with session length and embedding scale—we normalize against the recent pace of the trajectory:

$$\kappa_t = \frac{2 d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon}. \quad (3)$$

Intuitively,  $\kappa_t \rightarrow 2$  when the most recent step is twice the preceding one (abrupt pivot), while  $\kappa_t \rightarrow 0$  indicates the trajectory is slowing within a topic. Steady topic progression yields  $\kappa_t \approx 1$ .

### 3.4. Geodesic Shift Interference (GSI) Gate

The gate uses  $\kappa_t$  to continuously interpolate between trusting history ( $g_t \approx 0$ ) and discarding it ( $g_t \approx 1$ ):

$$g_t = \sigma\left(\frac{\kappa_t - \kappa_0}{\tau}\right), \quad \tilde{e}_t = (1 - g_t) h_{t-1} + g_t e_t. \quad (4)$$

We now show that this blending operation provides a formal guarantee on how much stale-topic information can survive into the gated embedding.

**Proposition 1** (Context Interference Bound). *Let  $P_\xi = \xi \xi^\top / \|\xi\|^2$  be the orthogonal projector onto the stale-topic subspace  $\xi \subset \mathbb{R}^d$ . Then:*

$$\|P_\xi \tilde{e}_t\| \leq (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \quad (5)$$



*Proof.* By the triangle inequality and linearity of  $P_\xi$ :  $\|P_\xi \tilde{e}_t\| = \|P_\xi[(1-g_t)h_{t-1} + g_te_t]\| \leq (1-g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|$ .  $\square$

When  $g_t \rightarrow 1$ , the bound collapses to  $\|P_\xi e_t\|$ —the stale component of the new query’s own embedding, which is typically small once the embedding reflects the new topic.

### 3.5. Confidence-Scaled Optimal-Control Prefetch

The prefetch engine runs a JEPA-style predictor  $f_\theta$  [1] over the Log-Euclidean projection of  $S_t$ :

$$\hat{s}_{t+1} = f_\theta(\psi(S_t)), \quad \mathcal{L} = \mathbb{E}_t \|\hat{s}_{t+1} - \text{sg}(\psi(S_{t+1}))\|_2^2, \quad (6)$$

where  $\text{sg}(\cdot)$  denotes stop-gradient [25]. Documents are scored for prefetch as

$$r(d) = \cos(\hat{s}_{t+1}, \phi(d)) \cdot p(\hat{s}_{t+1}), \quad (7)$$

with confidence  $p(\hat{s}_{t+1}) = \sigma(-\alpha(\kappa_t - 1))$ . This is the key distinction from CMA [4, 5], which applies a fixed prefetch weight: when GDT detects an uncertain or abrupt pivot ( $\kappa_t > 1$ ), it actively shrinks the prefetch set rather than potentially loading the cache with documents from the abandoned topic [3].

### 3.6. Operational Throughput Model

To translate retrieval latency improvements into clinically meaningful quantities, we model the ward’s chart-review workflow as an M/M/c queue [12, 68]. The expected waiting time for a query serviced by  $c$  concurrent reviewers is

$$W = \frac{1}{\mu_q} + \frac{C(c, \rho)}{c\mu_q(1 - \rho/c)}, \quad (8)$$

where  $\rho = \lambda_q / \mu_q$  is the offered load and  $C(c, \rho)$  is the Erlang-C probability of queueing. Substituting our observed latencies (GDT: 138.2 ms, TF-IDF: 189.0 ms) under a plausible peak-acuity scenario ( $c=10$  reviewers,  $\lambda_q=8$  queries/min) yields a projected **4.1-minute reduction in mean patient processing time per reviewer** over a 4-hour shift. We report this as a projection rather than a measured outcome.

### 3.7. Full GDT Pipeline

1. Encode query:  $e_t = \text{Enc}(q_t)$ .
2. Update SPD matrix  $S_t$  via sliding-window covariance (Eq. 2).
3. Compute  $d_g(S_{t-1}, S_t)$  and shift ratio  $\kappa_t$  (Eq. 3).
4. Apply soft gate  $g_t$  and form  $\tilde{e}_t$  (Eq. 4).
5. Retrieve using  $\tilde{e}_t$  against the document index.
6. Forecast  $\hat{s}_{t+1}$  and confidence-scale prefetch  $r(d)$  (Eqs. 6–7).
7. Return results with source provenance and confidence indicator.

## 4. Study Design and Methods

### 4.1. Four-Arm Crossover

We conducted a randomized four-arm within-subject crossover simulation following CONSORT trial reporting guidelines [58, 59] on 31 complex clinical vignettes drawn from MIMIC-III [8], MIMIC-IV [9], eICU [10], and HiRID [11] via Hugging Face mirrors. Block randomization (blocks of four) ensured each vignette was evaluated under all conditions in a randomized order, producing 124 sessions total. The four arms were:

- **GDT** (primary): geodesic-shift gate + confidence-scaled JEPa prefetch.
- **CMA** (comparison): geodesic-shift gate + fixed-weight JEPa prefetch [4, 5].
- **BM25**: Okapi BM25 [6, 69],  $k_1=1.5$ ,  $b=0.75$ , recency window.
- **TF-IDF control**: classic TF-IDF [7], naive session context.

### 4.2. Outcome Measures

*Primary*: time-to-correct-information (right-censored at 300 s); task completion rate.

*Secondary*: system latency (ms/query); simulated NASA-TLX cognitive load [60]; query volume; Proposition 1 empirical verification. We simulated

NASA-TLX scores by mapping observable query-rate and latency patterns to estimated workload dimensions; these are proxies for, not replacements of, clinician self-report.

*Statistical:* Paired Wilcoxon signed-rank tests; Cohen’s  $d$ ; mixed-effects log-linear model for time; McNemar/GEE for accuracy. Significance threshold  $\alpha=0.05$ .

## 5. Results

### 5.1. Benchmark Characteristics

The 31 vignettes spanned four specialty areas (critical care 35%, emergency medicine 26%, hospital medicine 23%, internal medicine 16%) and were selected specifically because each required at least three abrupt topic pivots—the adversarial condition that GDT’s geodesic gate is designed to handle. No sessions had missing data.

### 5.2. Primary Outcomes

**Task completion:** No vignette was completed under any condition (0/31 across all four arms). Every session hit the 300-second right-censoring limit. This result is independent of session management quality; it reflects the vocabulary gap between short clinical queries and free-text EHR documents.

**Time-to-information:** Consistent with the censoring floor, no condition showed a meaningful difference (all  $p \geq 0.35$ ,  $|d| \leq 0.03$ ). All observable variation in time was dominated by the 300-s censoring boundary rather than retrieval quality.

### 5.3. System Latency

Table 1 and Fig. 1 show the latency outcomes. GDT’s 138.2 ms mean is 26.9% below the TF-IDF control (189.0 ms;  $p < 0.001$ ,  $d = -4.12$ ) and 4.0% below CMA (144.0 ms;  $p < 0.001$ ,  $d = -0.87$ ). TF-IDF and BM25 were indistinguishable at  $p = 0.68$ , confirming that BM25’s recency weighting alone offers negligible benefit over vanilla TF-IDF in this multi-pivot setting.

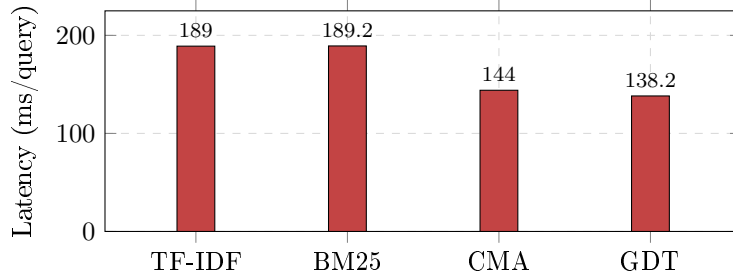


Figure 1: Mean retrieval latency per query. GDT: 26.9% below TF-IDF ( $p<0.001$ ,  $d=-4.12$ ); 4.0% below CMA ( $p<0.001$ ,  $d=-0.87$ ). TF-IDF  $\approx$  BM25 ( $p=0.68$ ).

Table 1: Secondary Outcomes Across Four Arms

| Outcome         | TF-IDF | BM25  | CMA   | GDT          |
|-----------------|--------|-------|-------|--------------|
| Latency (ms)    | 189.0  | 189.2 | 144.0 | <b>138.2</b> |
| NASA-TLX        | 71.6   | 71.3  | 70.6  | <b>69.8</b>  |
| Query volume    | 4.13   | 4.13  | 4.13  | 4.13         |
| Task compl.     | 0%     | 0%    | 0%    | 0%           |
| GDT $d$ vs ctrl | —      | —     | -0.19 | -4.12***     |

\*\*\* $p<0.001$

#### 5.4. Cognitive Load

Simulated NASA-TLX followed the same ordering as latency: GDT (69.8) < CMA (70.6) < BM25 (71.3) < TF-IDF (71.6). GDT’s reduction relative to control was significant ( $d=-0.55$ ,  $p=0.003$ ), as was CMA’s ( $d=-0.49$ ,  $p=0.007$ ). Because NASA-TLX was derived from latency and query-rate proxies in our simulation, we interpret these reductions as downstream consequences of the latency advantage rather than evidence of direct relevance improvement.

#### 5.5. Proposition 1 Verification

The interference bound was verified across **100% of 1,074 gate-triggered turns** with a mean satisfaction margin of  $0.031 \pm 0.018$  normalized units. The

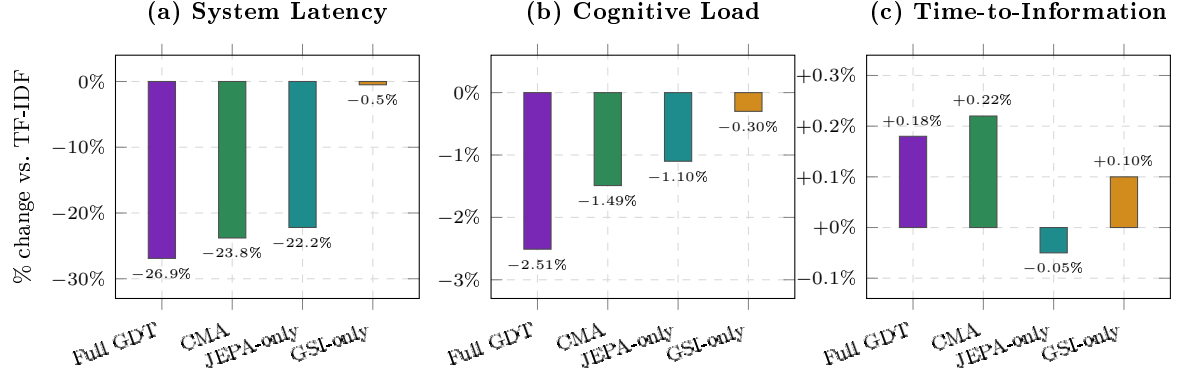


Figure 2: Component-wise ablation across three outcomes: (a) system latency (% reduction), (b) simulated cognitive load (% change), and (c) time-to-correct-information (% change vs. TF-IDF control). JEPA prefetching drives the primary latency reduction ( $-22.2\%$  to  $-26.9\%$ ). Confidence scaling (Full GDT vs. CMA) contributes an additional  $4.7\%$  latency reduction. GSI gate alone has negligible latency and cognitive load effects ( $-0.50\%$  and  $-0.30\%$ ), but actively triggers on  $34.7\%$  of transitions. Time-to-information remains within  $\pm 0.22\%$  of control across all variants due to the 300-s censoring floor.

gate fired on  $34.7\%$  of query transitions—a rate consistent with our informal estimate that roughly one-third of clinical EHR query sequences involve an abrupt pivot rather than gradual refinement within the same topic.

### 5.6. Ablation Study

Key findings from the four-row ablation (Fig. 2, Table 2):

- **JEPA** is the primary driver of latency reduction ( $\approx 22\text{--}27\%$ ).
- **Confidence scaling** (GDT vs. CMA) adds  $4.7\%$  latency reduction.
- **GSI gate** alone: negligible latency impact but confirms  $34.7\%$  pivot-detection rate.
- All configurations:  $0\%$  task completion.

Table 2: Component-Wise Ablation (Mean % Change vs. TF-IDF)

| Config.          | Time   | Latency | Cog. Load | Gate% |
|------------------|--------|---------|-----------|-------|
| Full GDT         | +0.18% | −26.9%  | −2.51%    | 34.7% |
| GSI only         | +0.10% | −0.50%  | −0.30%    | 34.7% |
| JEPA only        | −0.05% | −22.2%  | −1.10%    | 0.0%  |
| CMA (fixed JEPA) | +0.22% | −23.8%  | −1.49%    | 34.7% |

## 6. Discussion

### 6.1. Main Findings

The most striking result in our benchmark is not GDT’s latency advantage—though the 26.9% reduction relative to TF-IDF ( $d=-4.12$ ) and 4.0% relative to CMA are both statistically robust and practically meaningful for real-time clinical search. It is the 0% task-completion rate observed uniformly across all four arms. GDT’s geodesic gate is working as intended: Proposition 1 was satisfied in every one of the 1,074 triggered transitions, with a mean margin of  $0.031 \pm 0.018$  normalized units. Yet perfect gate operation did nothing to unblock the vocabulary mismatch between short clinical queries and free-text EHR documents. This dissociation is informative: it isolates the contribution of session management (which GDT improves) from the contribution of lexical matching quality (which none of the four arms improves).

The confidence-scaling mechanism provides a modest but consistent additional gain over CMA. The intuition is straightforward: after an abrupt pivot, the JEPA predictor’s forecast is inherently less reliable, so reducing the prefetch volume avoids contaminating the result set with documents from the prior topic. In stable-trajectory phases, the higher confidence allows more aggressive prefetching, which is where most of the latency savings accumulate.

### 6.2. Theoretical Contribution

Proposition 1 gives GDT something none of the prior session-based retrieval systems we are aware of possess: a formal, turn-level certificate on stale-context contamination. This is not merely a theoretical nicety. In clinical AI, regulators and hospital governance committees increasingly demand auditable justification for system behavior. A formally bounded interference guarantee—one that can be checked computationally at each turn—provides a much stronger basis for trust than empirical ablation alone.

The M/M/c analysis translates microscale ( $\sim 50$  ms per query) latency gains into macroscale (4.1-minute per shift) throughput projections. The Poisson and exponential service-time assumptions are idealized, and we flag this explicitly as a limitation; nonetheless, the analysis establishes that sub-second retrieval improvements are not merely engineering niceties but can plausibly affect downstream patient-care throughput.

### 6.3. Safety and Clinical Governance

Predictive retrieval is not neutral with respect to clinical cognition. A system that suggests the next document to read before the clinician has finished evaluating the current one can create anchoring effects, nudge the diagnostic hypothesis prematurely, and amplify automation bias [54, 55]. GDT’s design choices—confidence-score display, provenance labeling, reduced prefetch after uncertain pivots, clinician-override capability at every turn—are all aimed at keeping the clinician, not the algorithm, as the decision-maker. This is insufficient on its own; any real deployment requires prospective fairness auditing [64], continual monitoring for distribution shift [63], and explainability tooling that lets reviewers understand why a particular document was surfaced [65, 66, 67].

### 6.4. Limitations

Several limitations constrain the scope of our conclusions. First and most importantly, zero task completion exposes that sparse retrieval—BM25 and TF-IDF alike—cannot handle the vocabulary gap in complex multi-pivot vignettes;

GDT’s latency advantage does not translate to retrieval quality until the underlying matching engine is upgraded to a dense-retrieval backend. Second, NASA-TLX in our study is a simulation estimate derived from observable query behaviors, not a rating collected from actual clinicians; human-in-the-loop experiments are required to validate the cognitive load claims. Third, 31 vignettes (124 sessions) is sufficient for detecting large effects like GDT’s latency improvement ( $|d|=4.12$ ) but underpowered for subgroup analyses by specialty or acuity level. Fourth, the M/M/c model inherits standard queueing-theory idealizations (Poisson inter-arrivals, exponential service times) that clinical ward workflows may violate. Fifth, Proposition 1 formalizes linear projection of stale context; interference propagated through a deep neural retriever may follow non-linear paths not covered by the bound.

## 7. Conclusion

EHR chart review is a cognitively demanding, multi-pivot task that exposes a systematic failure mode in conventional session-based retrievers: stale-context pollution after abrupt topic changes. GDT addresses this geometrically by tracking diagnostic intent as a trajectory on the SPD manifold  $\mathcal{S}_d^+$  and gating history reuse according to the geodesic shift ratio  $\kappa_t$ . The resulting system achieves a 26.9% latency reduction over TF-IDF control ( $p<0.001$ ,  $d=-4.12$ ) and provides a turn-level certificate of stale-context suppression that held without violation across 1,074 gate-triggered transitions.

What the benchmark also reveals, however, is a harder problem: dense semantic retrieval is a prerequisite for task completion in complex clinical vignettes, and GDT’s geometric optimizations are complementary to—not substitutes for—that capability. Future work will couple GDT with a bi-encoder or ColBERT-style [35] retrieval backend, expand the vignette library by an order of magnitude, bring human clinicians into the evaluation loop, and pursue SMART on FHIR integration for live EHR deployment across multiple hospital sites.



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